

NETHOLABS RESEARCH

FUNCTIONAL BRAIN EMULATION

MULTI-MODAL NEURAL RECORDING DATA OVERVIEW

67 GB TOTAL DATA	42 RECORDINGS	9 SUBJECTS
11 MODALITIES	200+ CHANNELS	10 DEVICES

RECORDING DEVICES

DSI-24 EEG Wearable Sensing dry-electrode EEG headset. Wireless Bluetooth streaming for mobile recordings. Full 10-20 montage coverage. 24 CHANNELS 300 Hz SAMPLE RATE BT INTERFACE COM3 PORT	Kernel Flow 2 FNIRS
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TD-fNIRS helmet with 48 optical modules. Dual-wavelength (690nm Red + 850nm IR) hemoglobin measurement plus real-time signal quality tracking.

48	5 Hz	3	WS
MODULES	SAMPLE RATE	LSL STREAMS	KORTEX SDK

Gazepoint GP3

EYE TRACKING

Desktop eye tracker for lab-based recordings. Fixation position, pupil diameter, and validity tracking via TCP/XML protocol.

6	60 Hz	TCP	4242
CHANNELS	SAMPLE RATE	INTERFACE	PORT

Tobii Pro Glasses 3

MOBILE GAZE + IMU

Wearable eye-tracking glasses for mobile recordings. Binocular gaze, pupil diameter, and 9-axis IMU (accelerometer, gyroscope, magnetometer).

10	50 Hz	9	100 Hz
GAZE CH	GAZE RATE	IMU CH	IMU RATE

Pupil Labs Neon

MOBILE GAZE + IMU

Next-gen wearable eye tracker with scene camera. High-rate binocular gaze, 3D eye position, blinks, fixation detection, and onboard IMU.

17	200 Hz	10	110 Hz
GAZE CH	GAZE RATE	IMU CH	IMU RATE

Delsys Trigno

EMG

Wireless surface EMG system with auxiliary analog sensors. 16 muscle channels plus 48 accelerometer/auxiliary inputs via TCP SDK.

16	2000 Hz	48	148 Hz
EMG CH	EMG RATE	AUX CH	AUX RATE

StretchSense

HAND / BODY

Capacitive stretch sensor gloves and body suit. Full finger-level kinematics for both hands plus upper-body motion capture.

56

HAND CH

112

BODY CH

224

TOTAL CH

BLE

INTERFACE

Audio / Video Capture

A / V

Microphone audio stream, USB webcam (participant face), Pupil Neon scene camera (POV + iris), and two Raspberry Pi camera streams.

1

AUDIO CH

3

VIDEO FEEDS

2

PI CAMERAS

30 fps

FRAME RATE

LSL STREAM ARCHITECTURE

STREAM NAME	SOURCE DEVICE	CHANNELS	SAMPLE RATE	SIGNAL TYPE
DSI24	Wearable Sensing DSI-24	24	300 Hz	EEG (10-20 montage)
KernelFlow_Red	Kernel Flow 2	48	5 Hz	fNIRS Oxygenation (690nm)
KernelFlow_IR	Kernel Flow 2	48	5 Hz	fNIRS Deoxygenation (850nm)
KernelFlow_Quality	Kernel Flow 2	48	5 Hz	Signal Quality Index
Gazepoint	Gazepoint GP3	6	60 Hz	Fixation, Pupil, Validity
TobiiGlasses_Gaze	Tobii Pro Glasses 3	10	50 Hz	Binocular Gaze + Pupil
TobiiGlasses_IMU	Tobii Pro Glasses 3	9	100 Hz	Accel / Gyro / Mag

Stream Name	Source Device	Channels	Sample Rate	Signal Type
PupilNeon_Gaze	Pupil Labs Neon	17	200 Hz	Gaze, Blink, Fixation, 3D Eye
PupilNeon_IMU	Pupil Labs Neon	10	110 Hz	Accel / Gyro / Quaternion
Delsys_EMG	Delsys Trigno	16	2000 Hz	Surface Electromyography
Delsys_AUX	Delsys Trigno	48	148.1 Hz	Accelerometer / Auxiliary
StretchSense	StretchSense Gloves + Suit	224	Variable	Hand + Body Kinematics
Microphone	System Microphone	1	44100 Hz	Audio / Voice
StimulusMarkers	PsychoPy Stimulus System	Markers	Event	Stimulus Onset / Response
PiCamera / PiCamera2	Raspberry Pi 5	2	30 Hz	Frame Timestamps

DATA PIPELINE

SENSORS



Audio + Video



BRIDGE SCRIPTS

PYTHON
*_lsl.py
Device SDK → LSL



TRANSPORT

PROTOCOL
Lab Streaming Layer
Network-synced timestamps



STORAGE

LABRECORDER
.XDF files
BIDS-compliant naming



VISUALIZATION

REAL-TIME
Live Dashboard
PyQt5 + pyqtgraph

3D DEMO
Brain Emulation
Three.js + WebGL

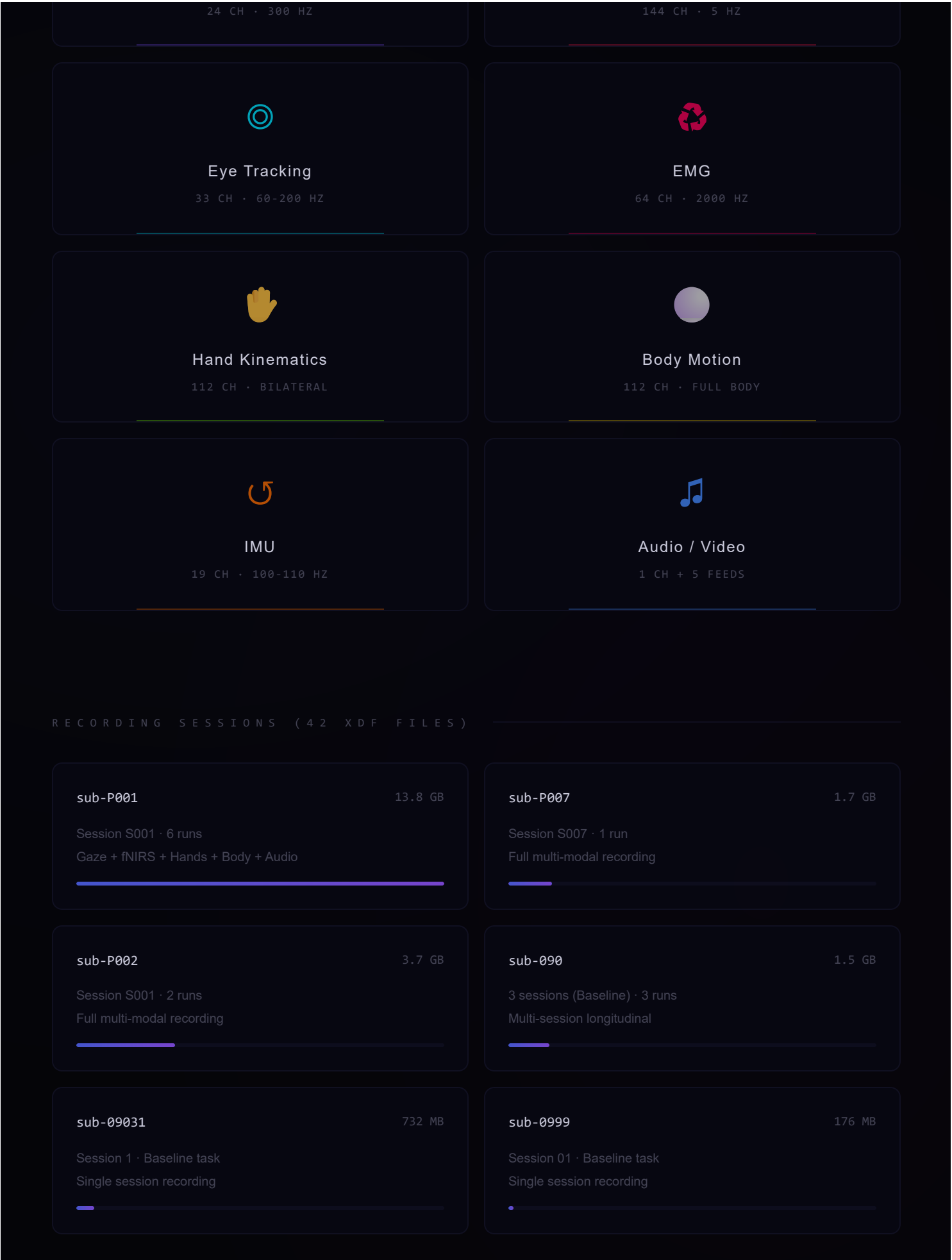
MODALITY COVERAGE



EEG



fNIRS



sub-2003

46 MB

2 sessions · Baseline task
Repeat participant



session_*

3.1 GB

14 timestamped sessions
Feb 27 – Mar 6, 2026



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